

NumPy Functions

Name	Keyword / Syntax	Description
Import NumPy	<code>import numpy as np</code>	Imports the NumPy library with the alias np or any name to your liking.
Create Array	<code>np.array()</code>	Creates a NumPy array from a list or tuple.
1D Array	<code>np.array([1,2,3])</code>	Creates a one-dimensional array.
2D Array	<code>np.array([[1,2],[3,4]])</code>	Creates a two-dimensional array (matrix).
3D Array	<code>np.array([[[...]]])</code>	Creates a three-dimensional array.
Zeros Array	<code>np.zeros(5)</code>	Creates an array filled with zeros.
Ones Array	<code>np.ones(5)</code>	Creates an array filled with ones.
Full Array	<code>np.full((2,3),100)</code>	Creates an array where every element is the given value.
Identity Matrix	<code>np.eye(4)</code>	Creates a square identity matrix (1s on diagonal).
Range Array	<code>np.arange(start, stop, step)</code>	Creates an array with evenly spaced values.
Evenly Spaced Values	<code>np.linspace(1,10,5)</code>	Creates a fixed number of equally spaced values between two numbers.
Shape	<code>arr.shape()</code>	Returns the number of rows and columns of an array.
Dimensions	<code>arr.ndim()</code>	Returns the number of dimensions of the array.
Size	<code>arr.size()</code>	Returns the total number of elements in the array.
Data Type	<code>arr.dtype()</code>	Returns the data type of the array elements.
Type Conversion	<code>arr.astype(int)</code>	Converts the array to another data type.

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Type Check	<code>type(arr)</code>	Returns the type of the object (numpy.ndarray).
Broadcasting	<code>Automatic</code>	Performs operations on arrays of different compatible shapes.
Mathematical Operations	<code>+, -, *, /, **</code>	Performs element-wise arithmetic operations.
Matrix Multiplication	<code>np.dot()</code> or <code>@</code>	Multiplies two matrices.
Install NumPy	<code>pip install numpy</code>	Installs the NumPy library.

Important AI Libraries

Library	Purpose
NumPy	Numerical computing and arrays
Pandas	Data analysis and manipulation
Scikit-learn	Machine learning algorithms
TensorFlow	Deep learning framework
PyTorch	Deep learning and neural networks
OpenCV	Computer vision and image processing